RANOpt A

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Applications & Use Cases for 6G RAN Product Development

AI-assisted engineering platform • 5G-Advanced baseline • Toward 3GPP Release 18 and 6G

6G product development needs a fast, explainable experimentation layer

Development challenge
RAN choices increasingly couple spectrum, duplexing, MIMO, carrier aggregation, propagation, scheduling and AI objectives.
Teams need rapid evidence before committing to detailed simulation, implementation or field trials.

Conventional workflow

Long setup and iteration cycles
Specialized tools and scripts
Results often separated from rationale

RANOpt AI workflow

Interactive what-if analysis
Traceable KPI calculation chain
AI recommendation with engineer review

Eight applications span product design, AI development and technical evaluation

Radio trade-space
Compare band, bandwidth, duplex, CA and MIMO choices.

Coverage–capacity
Expose morphology, ISD and loading trade-offs.

Scheduler design
Evaluate PF, RR, maximum-rate and AI-assisted policies.

KPI benchmarking
Compare baseline, target and AI-assisted outcomes.

AI sandbox
Define state, actions, rewards and inference behavior.

Model validation
Run golden scenarios, traces and consistency checks.

Operator adaptation
Prepare carrier data for offline training and testing.

Education & demos
Explain 5G-Advanced and AI-native RAN concepts visually.

A repeatable development loop connects radio assumptions to engineering decisions

MODEL

Configure the RAN
Band, duplex, bandwidth, MIMO, CA, morphology, users and scheduler.

EVALUATE

Calculate & compare
Path loss → SINR → CQI/MCS → spectral efficiency → throughput and KPIs.

IMPROVE

Recommend & validate
AI proposes actions; engineers inspect gains, trade-offs, traces and tests.

Use case 1 — De-risk 5G-Advanced radio product choices

For RAN Product Developers
Which radio configuration best meets the selected coverage, capacity, fairness and efficiency objectives towards 6G with AI implementation?

RF & propagation
Frequency, power, antenna gains, morphology, ISD and channel assumptions.

Spectrum & duplex
FDD/TDD layers, DL–UL allocation, bandwidth and component carriers.

MIMO & aggregation
gNB/UE layers, CA selection and effective spectral-efficiency gains.

Scheduler & KPIs
Throughput, 5th percentile, fairness, utilization, coverage and AI gain.

Use case 2 — MAC Scheduler Development
Explainable AI for RRM and MAC scheduling

OBSERVE

Network state
Load, users, SINR, CQI, PRBs, CA, MIMO and current KPIs.

ACT

Scheduling action
Scheduler choice, fairness weight, PRB priority and edge-user bias.

REWARD / EXPLAIN

Multi-KPI objective
Throughput, edge rate, fairness, utilization, delay and signaling—with rationale.

Use case 3 — Progress to Release 18 AI/ML lifecycle

TRAIN / TEST

Offline environment
Synthetic scenarios plus future operator datasets; train, test and benchmark policies.

PACKAGE

Controlled model release
Version the model, scenario coverage, KPI targets and validation evidence.

INFER / VERIFY

Browser decision support
Run inference, compare AI ON/OFF and retain human approval of recommendations.

Use case 4 — Build a safe 6G network digital twin

Development direction to Network Operators
Expand progressively, preserving the validated engineering core while adding network dynamics and AI control.

Multi-cell topology
Three-sector sites, macro/micro layers and geographic visualization.

Network dynamics
Mobility, handover, traffic evolution and inter-cell interference.

Coordinated control
Load balancing, scheduling, energy saving and predictive optimization.

Operational validation
Compare simulated actions with telemetry before network introduction.

3GPP Release 18 provides the baseline—RANOpt AI supplies the experimentation layer

Release 18 baseline
5G-Advanced expands AI/ML work across the NR air interface, NG-RAN and management.
RANOpt AI turns these themes into configurable studies; it does not claim 3GPP certification.

RAN1	NR air-interface AI/ML studies
RAN3	NG-RAN data collection & signaling
SA5	AI/ML training, testing, deployment & inference
RANOpt AI	Traceable what-if analysis and human approval

RANOpt AI is primarily a RAN product-development platform today

Positioning for RAN product developers
Its strongest present value is early product architecture, scheduler and AI algorithm development, KPI trade-off analysis, validation and customer demonstrations.
Operator value expands as real carrier data and multi-cell network behavior are added.

Best fit today

Radio configuration trade-space
AI-RRM and scheduler prototyping
Repeatable validation and demonstrations

Operator evolution

Carrier KPI and configuration-data ingestion
Multi-cell interference, mobility and handover
Operator-specific PPO and telemetry validation

DEVELOPMENT- Motivation
to RAN Product Developers and Wireless
Carriers

Build the AI and the RAN together—without losing engineering control

Start with repeatable cell-level studies •
Add offline PPO and operator data • Scale
to multi-cell 6G digital-twin validation